

AE 21: Inference for Logistic Regression Models

! Important

- Open [RStudio](#) and create a subfolder in your AE folder called “AE-21”.
- Go to the [Canvas](#) and locate your AE-21 assignment to get started.
- Upload the `ae-21.qmd` and `framingham.csv` files into the folder you just created. The `.qmd` and PDF responses are due in Canvas. You can check the due date on the Canvas assignment.

Packages

```
library(tidyverse)
library(broom)
library(ggformula)
library(mosaic)
library(knitr)

heart_disease <- read_csv("framingham.csv") |>
  select(TenYearCHD, totChol, currentSmoker, BMI, BPMeds, cigsPerDay, prevalentHyp, heartRate)
drop_na()
```

Data: Framingham study

This data set is from an ongoing cardiovascular study on residents of the town of Framingham, Massachusetts. We want to predict if a randomly selected adult is high risk for heart disease in the next 10 years.

Response variable

- `TenYearCHD`:
 - 1: Patient developed heart disease within 10 years of exam
 - 0: Patient did not develop heart disease within 10 years of exam

What are my predictor variables?

Based on your group, use the following as your predictor variables.

- Group 1:
 - `totChol`: total cholesterol (mg/dL)
 - `currentSmoker`: 1 if current smoker, 0 otherwise
- Group 2:
 - BMI: patient's body mass index
 - `BPMeds`: 1 if they are on blood pressure medication, 0 otherwise
- Group 3:
 - `cigsPerDay`: number of cigarettes patient smokes per day
 - `prevalentHyp`: 1 if patient was hypertensive, 0 otherwise
- Group 4
 - `sysBP`: systolic blood pressure (mmHg)
 - `diabetes`: 1 if patient has diabetes, 0 otherwise

Exercise 0

Fit two logistic regression models predicting `TenYearCHD`, one for each of your explanatory variables. Keep track of which one has a quantitative and which has a categorical predictor.

Exercise 1

For your quantitative predictor, conduct a hypothesis test to determine whether the slope of your variable is statistically significant. On the white board:

1. Specify the null and alternative hypotheses.
2. Compute the test statistic.
3. Give the p-value.
4. Interpret the result.

Exercise 2

For your quantitative variable. Construct a 99% confidence interval. Hint, you can use the `conf.level` argument in `tidy` to change the confidence level. On the white board:

1. Interpret your confidence interval in *log-odds*.
2. Interpret your confidence interval in *odds*.

Exercise 3

For your categorical predictor, conduct a hypothesis test and construct 95% confidence interval for your β_1 . Interpret these in context and be prepared to discuss with the class. Note that you need not write down the entire hypothesis testing framework.

Exercise 4

Why do you think it doesn't quite make sense to talk about prediction intervals or confidence intervals in the context of a logistic regression model?

Submission

! Important

To submit the AE:

- Render the document to produce the PDF with all of your work from today's class.
- Upload your QMD and PDF files to the Canvas assignment.